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Azure Deployment Cost & Architecture Report

May 2026 · Estimated Monthly Costs

1. Architecture Summary

This deployment uses a straightforward three-tier web application architecture hosted entirely on Microsoft Azure. The stack is optimized for small-to-medium workloads with a focus on managed services to minimize operational overhead.

Deployed Resources

- Azure App Service (B2 Linux) — Hosts the web application. Provides built-in load balancing, auto-restart, and deployment slots for zero-downtime releases.
- Azure SQL Database (Basic tier) — Fully managed relational database with 5 DTUs and 2 GB of storage. Handles transactional data with automated backups and geo-redundancy options.
- Azure Blob Storage (Hot LRS) — Object storage for user uploads, static assets, and application logs. Locally redundant, with lifecycle policies to tier aging data.
- Azure Virtual Network — Isolates resources in a private network. The App Service uses VNet Integration; the SQL Database uses a private endpoint to prevent public internet exposure.
- Azure Monitor + Log Analytics — Centralizes metrics, traces, and logs. Configured with alert rules for CPU > 85%, failed requests, and database DTU saturation.
- Azure Key Vault — Stores connection strings, API keys, and certificates. App Service retrieves secrets via Managed Identity (no credentials in code).

2. Itemized Cost Breakdown

Estimates are based on the Azure Pricing Calculator (Pay-As-You-Go, East US region, May 2026). Actual costs may vary with usage.

Resource	Configuration	Est. Monthly Cost
Storage (Premium Managed Disks) SSD	30 GB	\$2.01
Azure Virtual Network	Standard IPv4 Static Public IP	\$1.29





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The following strategies can meaningfully reduce the monthly bill without sacrificing reliability or performance:

Strategy	Detail & Estimated Savings
Reserved Instances	Commit to a 1-year or 3-year App Service plan for up to 40% savings vs. pay-as-you-go. Ideal when workloads are predictable.
Auto-Scaling / Off-Hours Shutdown	Scale App Service down to B1 tier nights/weekends, or use Azure Automation to stop dev/test instances. Can cut compute costs by ~30–50%.
Azure Free Tier & Dev/Test Pricing	Use Azure Dev/Test subscriptions for non-production environments — up to 55% off Windows VMs and App Service. SQL Basic tier is already near the free limit.
Blob Storage Lifecycle Policies	Move infrequently accessed blobs from Hot → Cool → Archive tiers automatically. Archive is ~90% cheaper than Hot for rarely read data.

Quick Wins (Implement Immediately)

- Enable Azure Advisor — free service that automatically flags idle resources, unattached disks, and right-sizing opportunities.
- Tag all resources with environment=prod / env=dev to enable cost filtering in Cost Management + Billing dashboards.
- Set a budget alert at \$5/mo in Azure Cost Management to catch unexpected spikes before they compound.

By combining Reserved Instances with off-hours scaling, it is realistic to reduce the estimated \$3.3/mo to approximately \$2-5/mo — a saving of 30–45% — with no architectural changes required.